

EXP 4 Design of Low Pass Filter using Windowing Techniques

4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique in MATLAB

```
clc;
clear;
close all;

%% --- Input Parameters ---

fp = 1000;    % Passband cutoff (Hz)
fs = 4000;    % Sampling frequency (Hz)
N = 50;       % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm);

%% --- Plot Frequency Responses ---

figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
```

OUTPUT

```
--- Hamming Window Coefficients ---
0.0010
0
-0.0013
0
0.0021
0
-0.0034
0
0.0055
0
-0.0084
0
0.0125
```

0
-0.0181
0
0.0260
0
-0.0379
0
0.0580
0
-0.1026
0
0.3168
0.4995
0.3168
0
-0.1026
0
0.0580
0
-0.0379
0
0.0260
0
-0.0181
0
0.0125
0
-0.0084
0
0.0055
0
-0.0034
0
0.0021
0
-0.0013
0
0.0010

